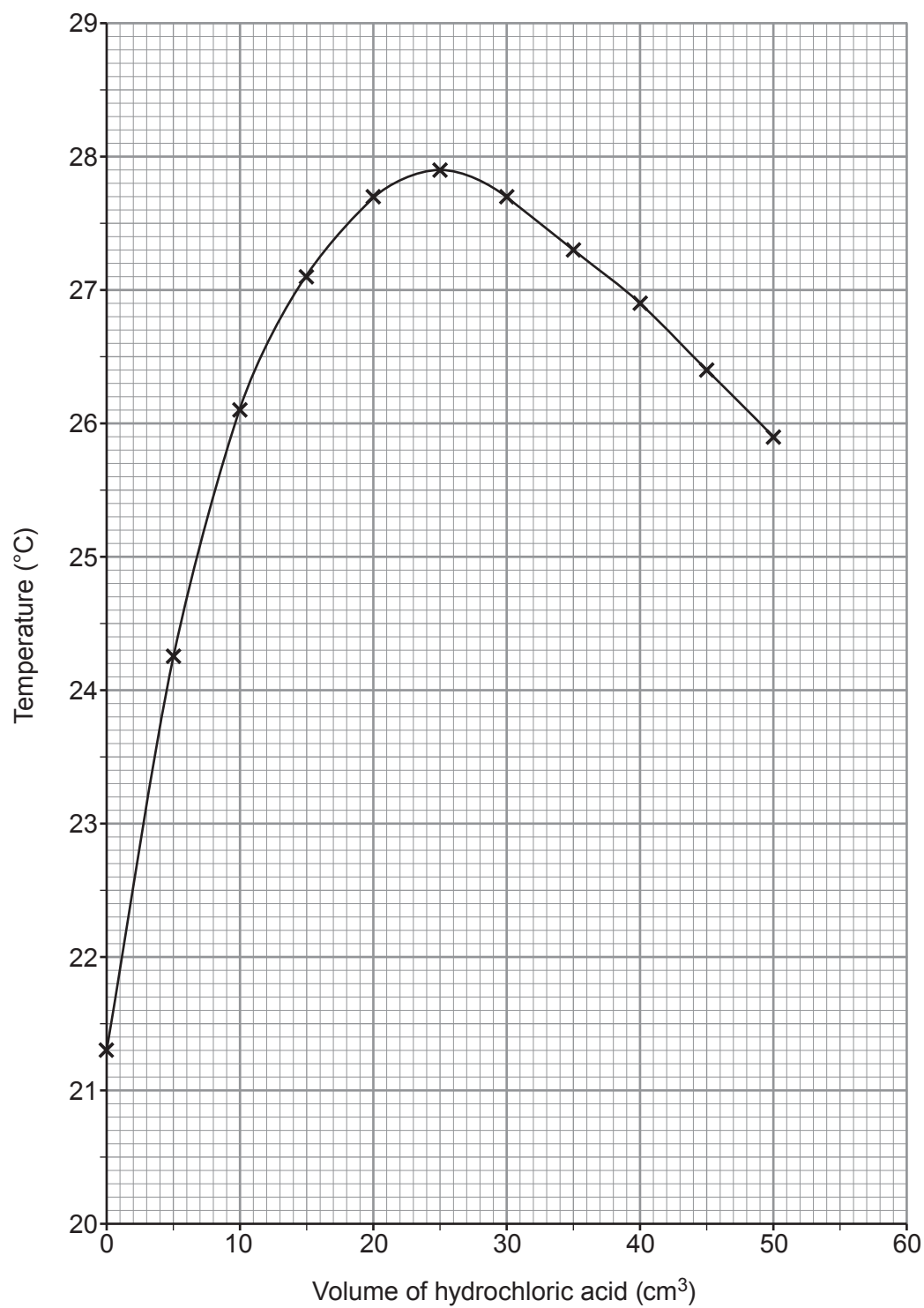


7. (a) The graph shows how the temperature changes when dilute hydrochloric acid is added gradually to sodium hydroxide solution.



The pH of the solution also changes as the hydrochloric acid is added to the sodium hydroxide.

Explain the temperature changes shown on the graph and relate these to the change in pH during the reaction. You do **not** need to include equations in your answer. [6 QER]

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- (b) The reaction between hydrochloric acid and sodium hydroxide produces sodium chloride and water.

Give the **ionic** equation to show how water is formed during this reaction. [1]

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END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1

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3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2									
7 Li Lithium 3		9 Be Beryllium 4								19 F Fluorine 9		20 Ne Neon 10							
23 Na Sodium 11		24 Mg Magnesium 12								35.5 Cl Chlorine 17		40 Ar Argon 18							
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

24

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